

L6: Transforming Geocentric Coordinates to Official Coordinates

RF: Realisations of RS

- Implemented by providing coordinates at number of dedicated points
- Examples:
 - **ITRF_{yy}**
 - ITRF in year yy
 - Accuracy 5-15mm and 2-3mm/year
 - Maintained by Scientific Community
 - **ETRS89**
 - European Terrestrial RF
 - Frozen
 - **Definition of official coordinates in Germany**
 - **WGS84**
 - World Geodetic System
 - US Military System
 - Also specifies reference ellipsoid, set of fundamental constant and Earth gravity model
 - Adapted to ITRF - Differences in ~dm
 - Definition of GPS Satellites

Similarity Transformation between CS

$$\begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix} = \lambda \mathbf{R}_3(\varepsilon_z) \mathbf{R}_2(\varepsilon_y) \mathbf{R}_1(\varepsilon_x) \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

But Small rotations: Approximations valid

$$\begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix} = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

Conversion from Ellipsoidal Coordinates

- Ellipsoidal coordinates easier to interpret on earth
- Ellipsoidal coordinates need definition of reference ellipsoid

- λ ... longitude, φ ... latitude, h ... ellipsoidal height
- $\lambda, \varphi, h \rightarrow X, Y, Z$

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} (N(\varphi) + h) \cos \varphi \cos \lambda \\ (N(\varphi) + h) \cos \varphi \sin \lambda \\ \left(\frac{b^2}{a^2} N(\varphi) + h\right) \cdot \varphi \end{bmatrix}$$

Transverse radius of curvature

$$N(\varphi) = \frac{a^2}{\sqrt{a^2 \cdot \cos^2 \varphi + b^2 \cdot \sin^2 \varphi}}$$

Conversion to Ellipsoidal Coordinates

$$\begin{bmatrix} \varphi \\ \lambda \\ h \end{bmatrix} = \begin{bmatrix} \tan^{-1} \left(\frac{Z}{\sqrt{X^2 + Y^2}} \cdot \left(1 - \frac{a^2 - b^2}{a^2} \cdot \frac{N(\varphi)}{N(\varphi) + h} \right)^{-1} \right) \\ \tan^{-1} \left(\frac{Y}{X} \right) \\ \frac{\sqrt{X^2 + Y^2}}{\cos \varphi} - N(\varphi) \end{bmatrix}$$

Iterate

$$N(\varphi) = \frac{a^2}{\sqrt{a^2 \cdot \cos^2 \varphi + b^2 \cdot \sin^2 \varphi}}$$

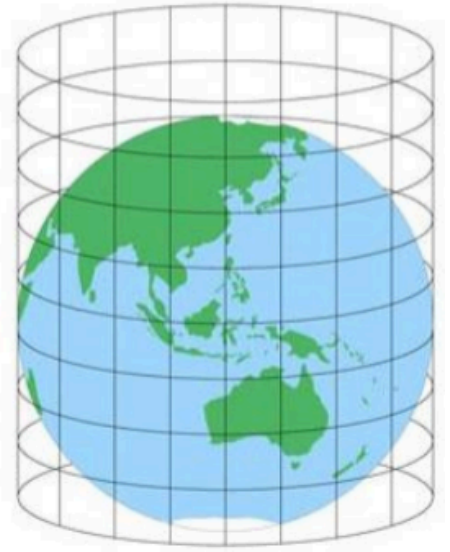
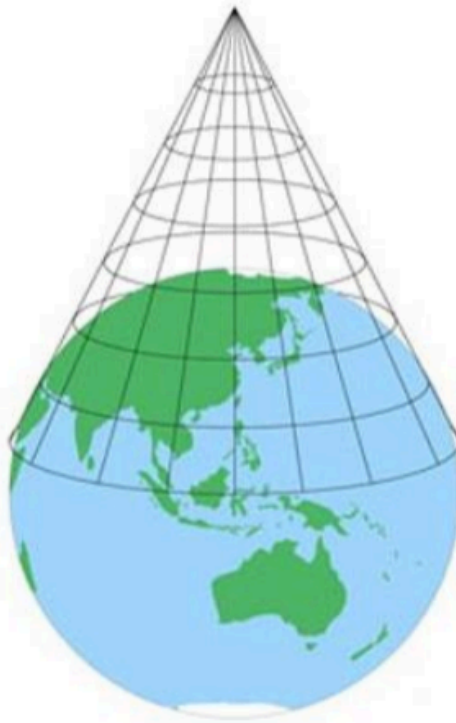
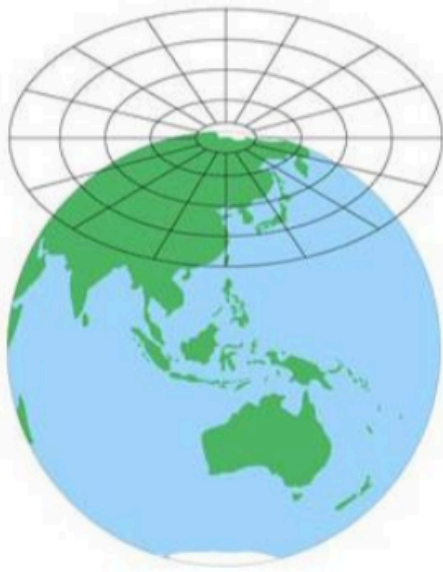
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- Official coordinates separate by position(2D) and height
 - Position needs to be **metric** $\Rightarrow$ no longitude and latitude
 - Height needs to be **physical** $\Rightarrow$ no ellipsoidal height
 - Official coordinates need to be stable (no change due to earth process (-> No ITRF/WGS84 for Europe))

Map Projections

azimuthal

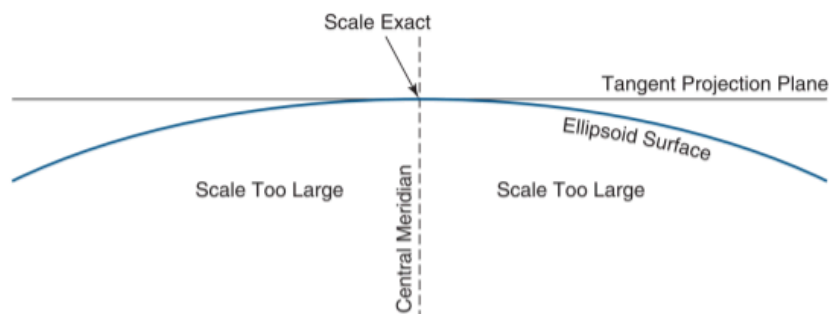
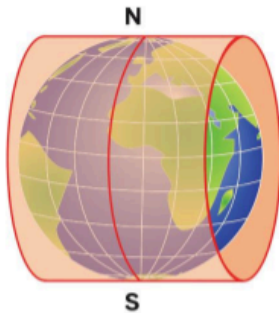
conical

cylindrical



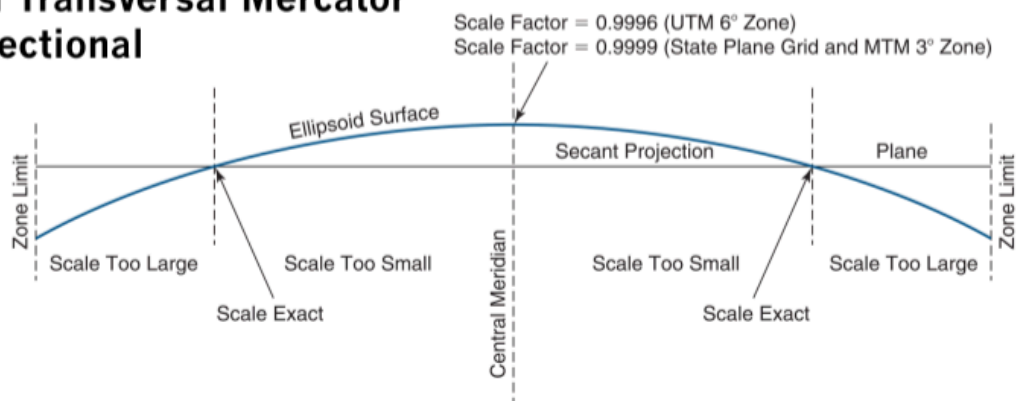
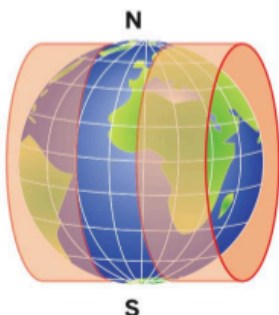
• Cylindrical Map Projection

- Rectangular, planar coordinate system
- East axis, North axis



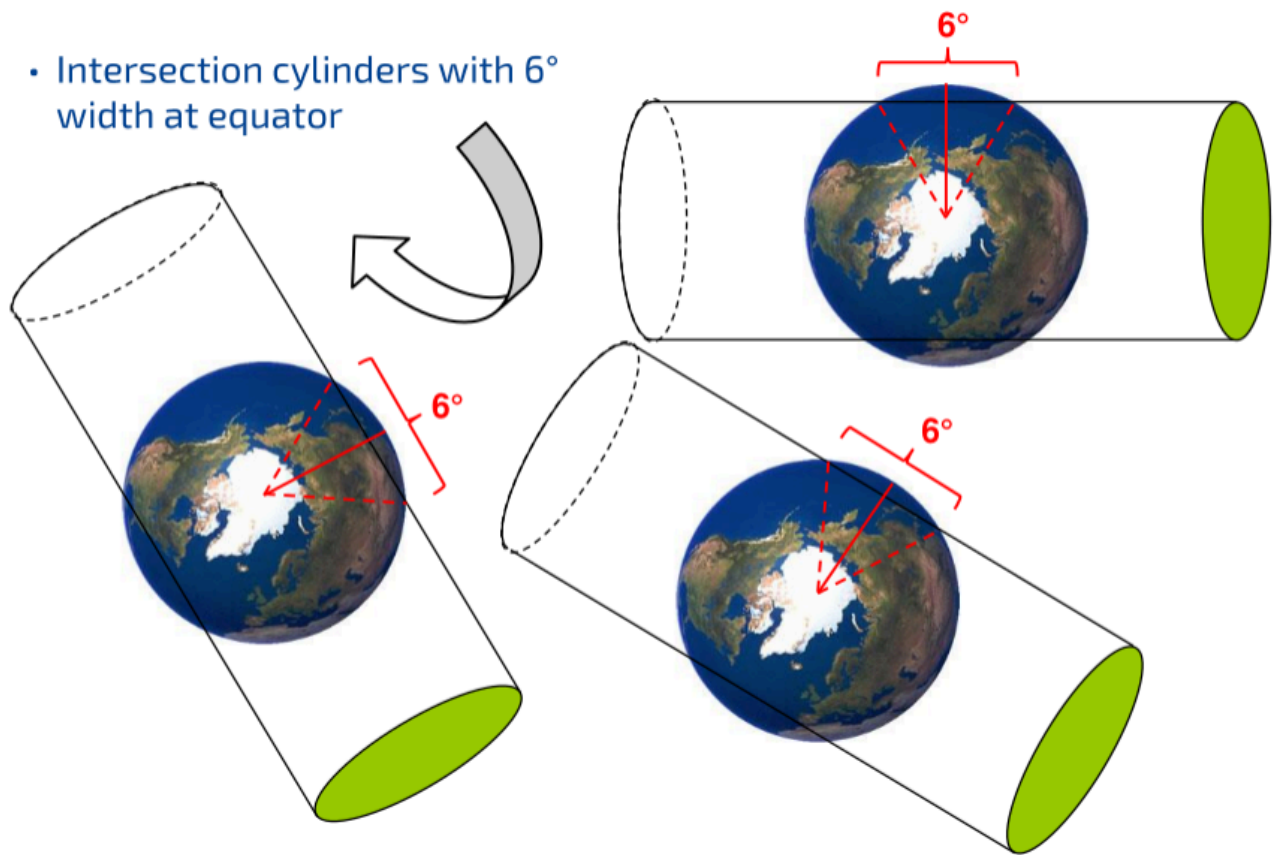
Above: Gauß-Krüger => tangential

Below: **Universal Transversal Mercator (UTM) => intersectional**



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- UTM:

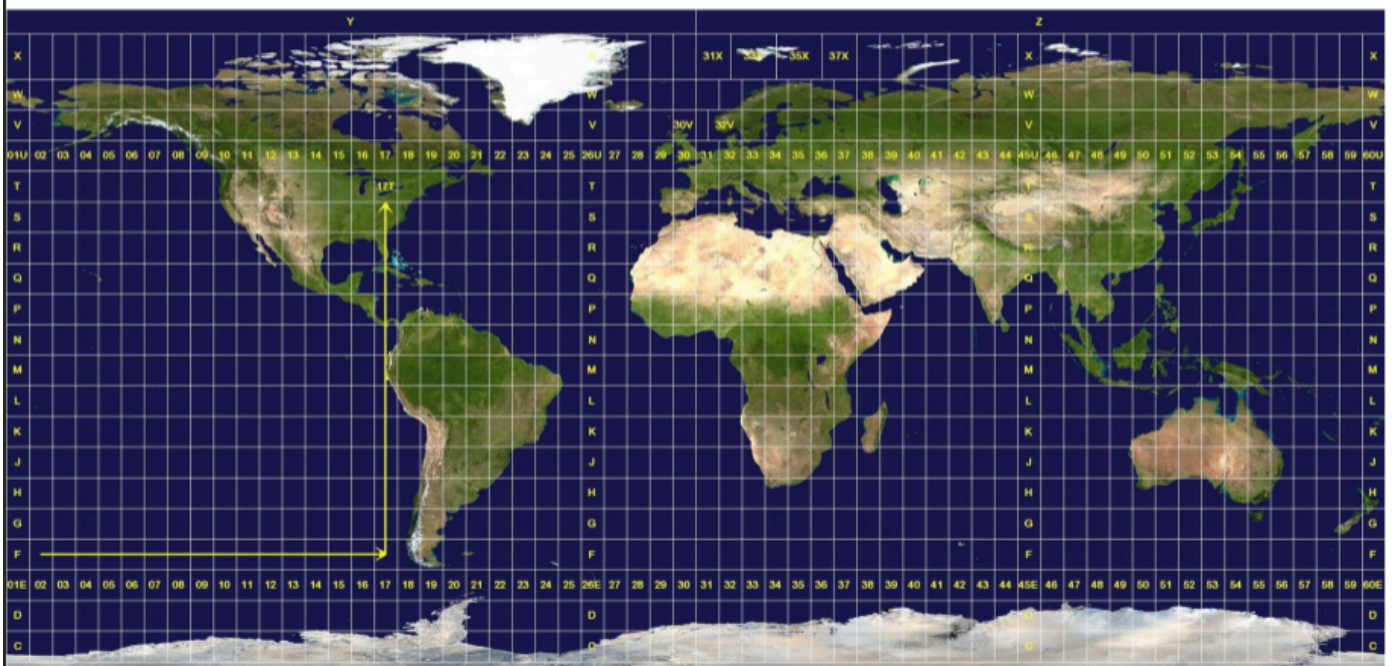
- Intersection cylinders with 6° width at equator



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- 6° wide Zones
- **Isogonal /Equal Angles**
- Northing value of the equator: $N = 0m$
- Easting value of each entral meridian: $E = 500'000m$
- Scale Factor at central meridian: 0.9996
- Projection limits of Latitude 80°S and 80°N

- 60 stripes of 6° each = 360°



- Germany: stripes 31, 32, 33
- North Rhine-Westphalia: only stripe 32 with central meridian at 9°

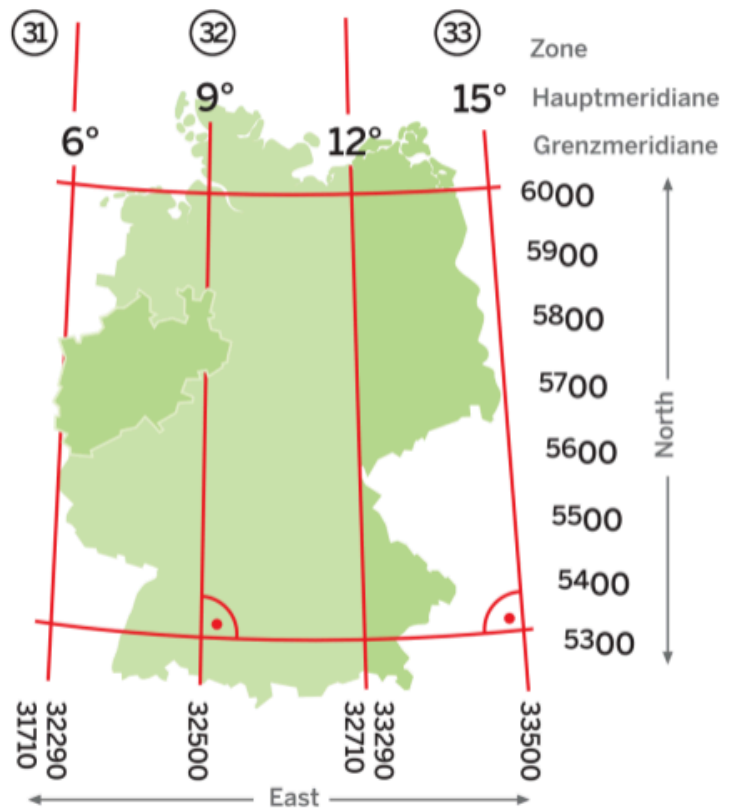
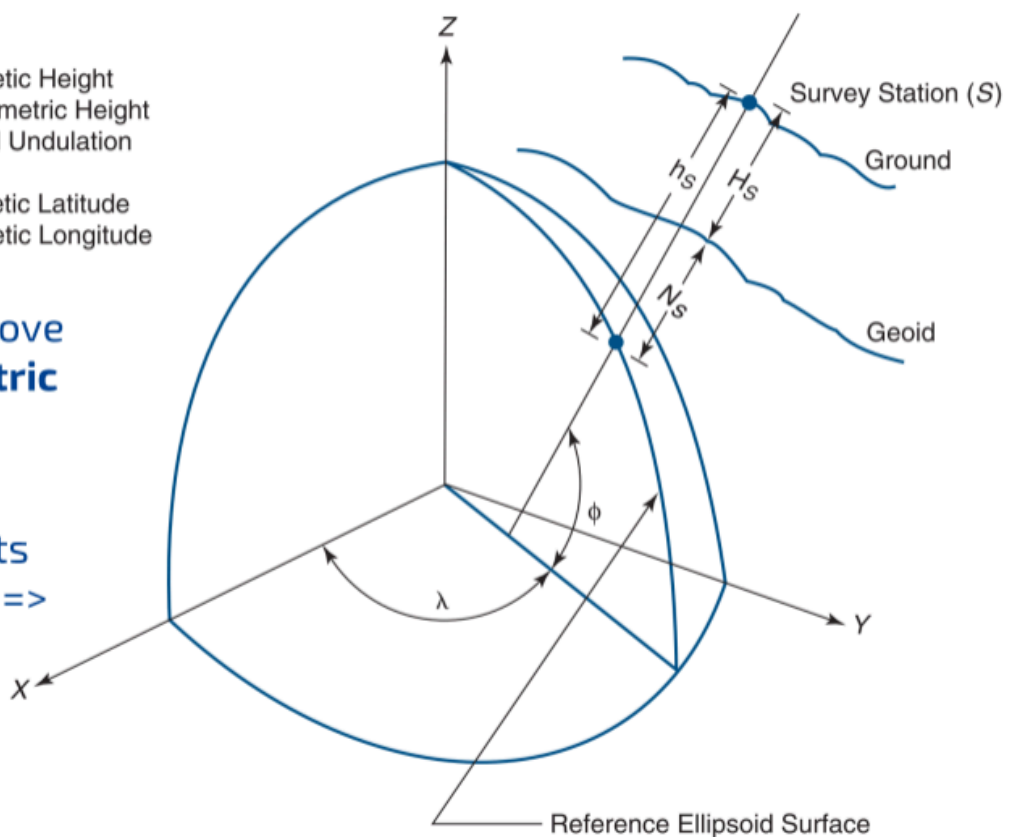


Abb. 6: Die Lage von NRW in der UTM-Zone 32

Physical Heights

h = Geodetic Height
 H = Orthometric Height
 N = Geoid Undulation
 $h = H + N$
 ϕ = Geodetic Latitude
 λ = Geodetic Longitude

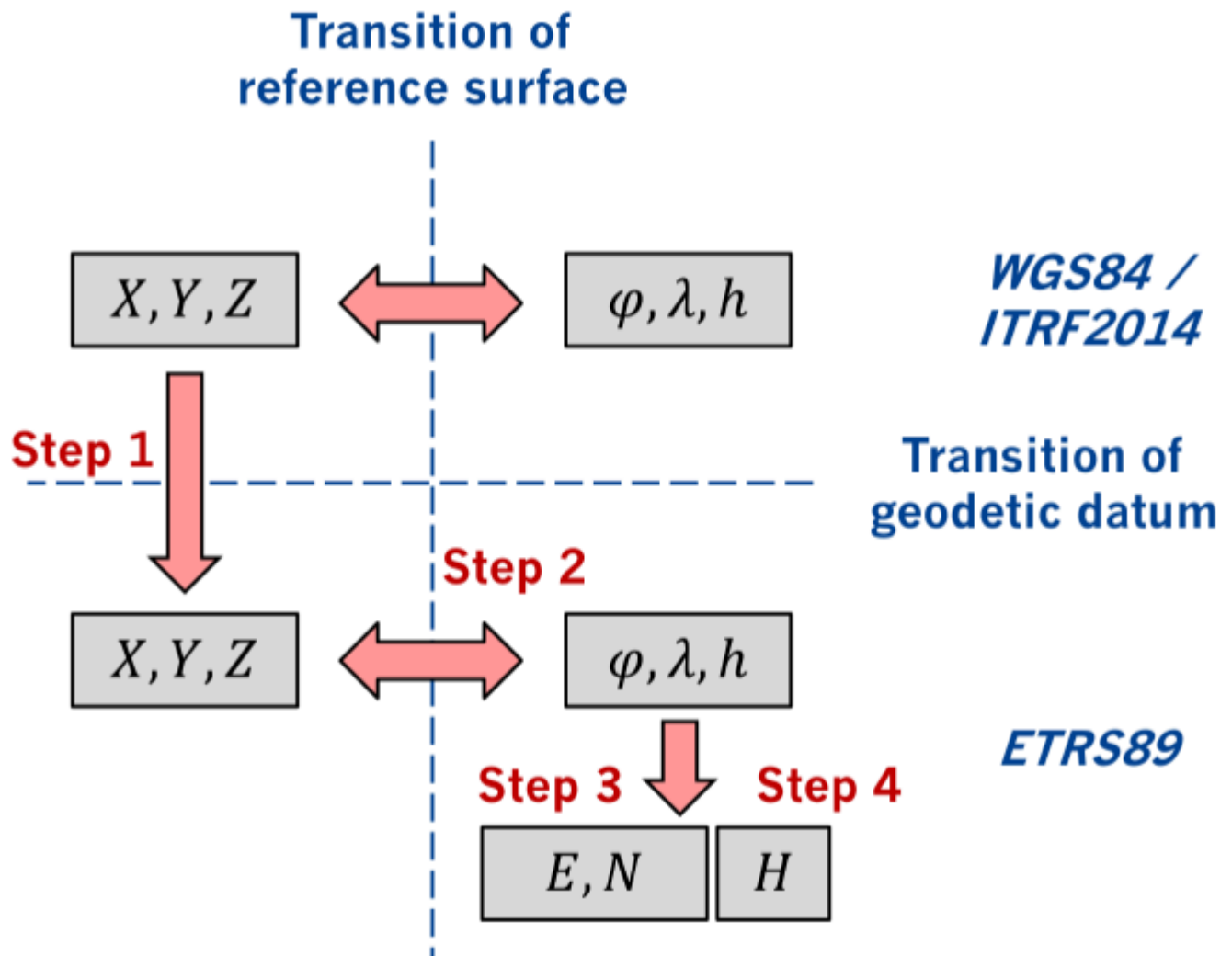
- Physical height above geoid => **orthometric heights**
- In Germany: heights above quasi geoid => **normal heights**



source:
 avanagh
 (2014)

- Difference between geoid and ellipsoid: Geoid undulation
 - Needs to be known to get physical heights out of ellipsoidal heights

Transformation to official coordinates



1. Transform WGS84 $\rightarrow$ ETRS89

- **Transition of Geodetic Datum**
- Baseline $\Delta X, \Delta Y, \Delta Z$ given in WGS84
- Apply pure translation

2. Cartesian $\xrightarrow{GRS80}$ Ellipsoidal

- **Transition of reference surface**
- Use Conversion from Cartesian to Ellipsoid (GRS80)

3. Transform latitude and longitude to UTM coordinates

- **Projection**

- $E = E_0 + [1] \cdot \Delta L + [3] \cdot \Delta L^3 + [5] \cdot \Delta L^5$
- $N = m \cdot G + [2] \cdot \Delta L^2 + [4] \cdot \Delta L^4 + [6] \cdot \Delta L^6$
- $\Delta L = L - L_0$, with $L_0 = 9^\circ$ for North Rhine-Westfalia
- $E_0 = \left(\frac{L_0 + 3^\circ}{6^\circ} + 30.5 \right) \cdot 10^6$
- $m = 0.9996$
- $G = G_0 \cdot B + G_2 \cdot \sin 2B + G_4 \cdot \sin 4B + G_6 \cdot \sin 6B$
- $[1] = \frac{m}{\rho} \cdot \bar{N} \cdot \cos B$
- $[3] = \frac{m}{6\rho^3} \cdot \bar{N} \cdot \cos^3 B \cdot (1 - t^2 + \eta^2)$
- $[5] = \frac{m}{120\rho^5} \cdot \bar{N} \cdot \cos^5 B \cdot (5 - 18t^2 + t^4 + \eta^2 \cdot (14 - 58t^2))$
- $[2] = \frac{m}{2\rho^2} \cdot \bar{N} \cdot \cos^2 B \cdot t$
- $[4] = \frac{m}{24\rho^4} \cdot \bar{N} \cdot \cos^4 B \cdot t \cdot (5 - t^2 + 9 \cdot \eta^2)$
- $[6] = \frac{m}{720\rho^6} \cdot \bar{N} \cdot \cos^6 B \cdot t \cdot (61 - 58t^2 + t^4)$

- $\rho = \frac{180^\circ}{\pi}$
- $\bar{N} = \frac{c}{\sqrt{1+\eta^2}}$
- $t = \tan B$
- $\eta^2 = \frac{a^2 - b^2}{b^2} \cdot \cos^2 B$
- $c = \frac{a^2}{b}$

$\lambda = L = \text{Longitude}$

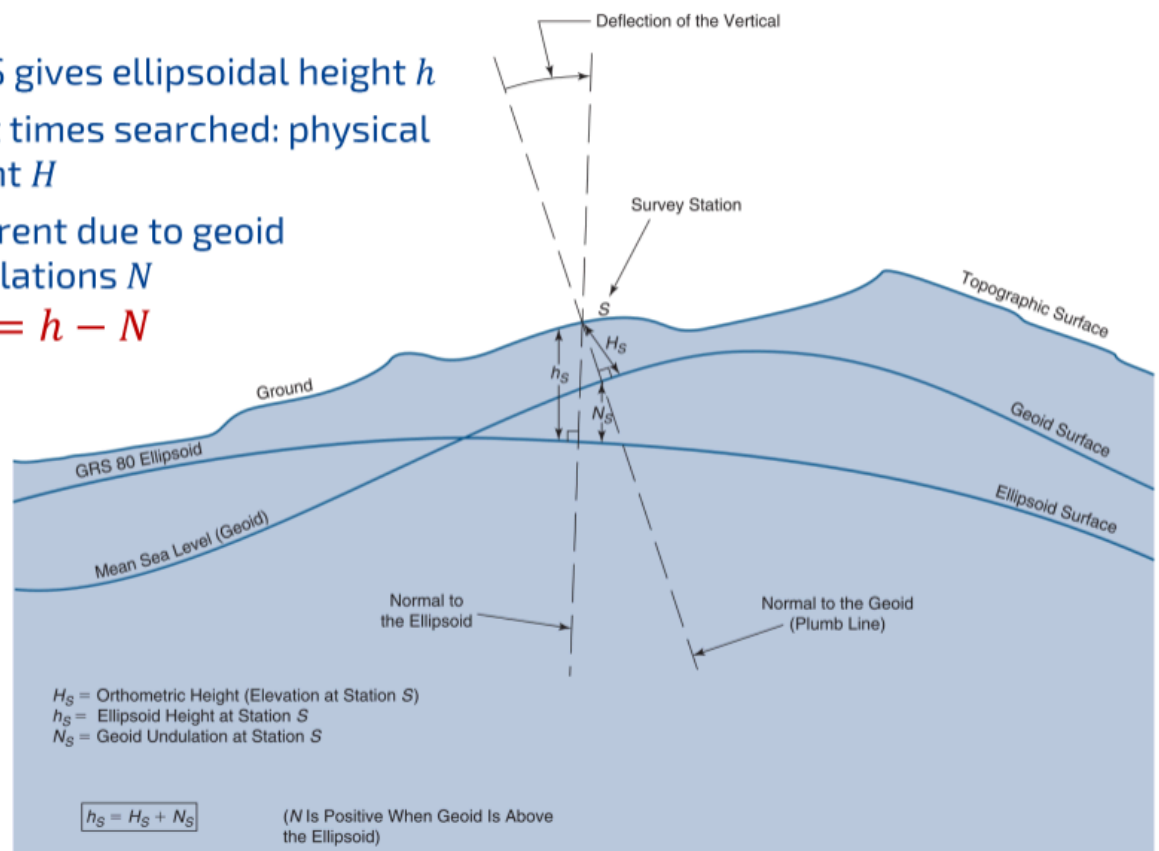
$\varphi = B = \text{Latitude}$

Parameter	GRS80 ellipsoid
G_0	111'132.952547 m/°
G_2	-16'038.5088 m
G_4	16.8326 m
G_6	-0.0220 m
a	6'378'137.000 m
b	6'356'752.314 m

4. Ellipsoidal height $\xrightarrow{GCG16}$ physical height

- Transition to referene surface
- GCG2016 quasi geoid undulations

- GNSS gives ellipsoidal height h
 - Most times searched: physical height H
 - Different due to geoid undulations N
- $\Rightarrow H = h - N$



Source:
anaght
14)